Sahaj Bhadja

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Summary

Research Software Engineer with 2+ years building **Python applications**, **scientific data pipelines**, cloud-backed services, and **reproducible ML workflows** across Linux, databases, containers, testing, and monitoring.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **scientific data pipelines**, implementing validation, reconciliation, error handling, cloud storage, incremental processing, CI/CD releases, and stable production monitoring workflows.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Maintained 90%+ scoring consistency across 10K+ submissions through **structured model evaluation**, model switching, fallback paths, persistent results, and reviewer controls supporting reproducible analytical workflows reliably for technical users.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **research data pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst - Research Fellowship | *India*
- Achieved 96%+ classification accuracy across 650K+ URLs by comparing **Random Forest, CNN, RNN, LSTM, Bi-LSTM, and GRU models** within a reproducible research evaluation pipeline for malware detection.
- Engineered 21 behavioral indicators covering URL length, IP presence, and suspicious keywords, translating noisy inputs into **classification features** that exposed phishing and malware patterns reliably at scale.

Skills

Languages: Python, Bash, Java, TypeScript, SQL
Applications & APIs: FastAPI, Flask, React, Node.js, REST APIs, Microservices
Data & Storage: PostgreSQL, MySQL, MongoDB, SQL Server, AWS RDS, Data Pipelines, ETL, Data Modeling, Cloud Storage
Research & ML: PyTorch, Transformers, Machine Learning, Deep Learning, Model Evaluation, Feature Engineering, Data Validation
Systems & Cloud: Linux, Windows, Git, GitHub, Docker, Kubernetes, AWS
Quality & Operations: CI/CD, Unit Testing, Integration Testing, Debugging, Monitoring, Technical Documentation

Projects

Scientific Core Workflow Orchestrator - Design Concept
- Designed a **scientific workflow orchestrator** concept in Python with Linux command-line utilities, containerized jobs, Kubernetes scheduling, cloud storage, validation checkpoints, health monitoring, failure recovery, and researcher-facing documentation.

Chat2SQL | Live
- Built an **AI-powered SQL workspace** across Java and TypeScript, supporting 10+ databases with schema exploration, query generation, Docker, Maven, Node.js, and cross-platform development workflows for efficient data management.

Big-Brain | Live
- Built an **AI-powered research data platform** with secure authentication, real-time collaboration, semantic search, and ingestion across 50K+ documents, reducing review time by 60% through scalable automated workflows.